Lecture 2: Eigenvalues & Spectral Mapping

Core Definitions

In this session, we analyze how linear transformations scale specific vector subspaces.

Definition : Eigenvalues and Eigenvectors

Let $T : V \to V$ be a linear operator on a vector space V over a field F. A non-zero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in F$ such that:

$$T(v) = \lambda v$$

The scalar λ is called the **eigenvalue** corresponding to the eigenvector v.

Fundamental Theorems

Theorem : Linear Independence of Eigenvectors

Let $T : V \to V$ be a linear operator. If $v_1, v_2, \ldots, v_m$ are eigenvectors corresponding to **distinct** eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$, then the set $\{v_1, v_2, \ldots, v_m\}$ is linearly independent.

Theorem : Spectral Mapping Theorem (Polynomials)

Let $T : V \to V$ be a linear operator with an eigenvalue λ. For any polynomial $p(t) = a_n t^n + \ldots + a_1 t + a_0$, $p(\lambda)$ is an eigenvalue of the operator $p(T)$, operating as:

$$p(T)(v) = p(\lambda)v$$

Homework Practice

Exercise : Characteristic Polynomial Evaluation

Consider the 2×2 matrix A representing a linear transformation on $\mathbb{R}^2$:

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

1. Compute the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.
2. Find the distinct eigenvalues λ_1 and λ_2 by solving:

$$\lambda^2 - 7\lambda + 10 = 0$$